

Question : Create a Dockerfile for an Express.js Application

Objective: Write a simple Dockerfile to containerize an Express.js application.

Task:

1. Create a directory named `express-app` on your host machine.
2. Inside the directory, create a simple Express.js application:
 - Create `package.json` with Express dependency.
 - Create `server.js` with a basic web server.
3. Write a **Dockerfile** that:
 - Uses `node:18-alpine` as the base image (lightweight).
 - Sets the working directory to `/app`.
 - Copies `package.json` and installs dependencies.
 - Copies the application code.
 - Exposes port `3000`.
 - Runs the Express app using `node server.js`.
4. **Build** the image with the tag `express-app:1.0`.
5. **Run** a container from the image, mapping port `3000` on your host to port `3000` in the container.
6. Access the API at `http://localhost:3000` and verify it works.

Sample Files for Students:

`package.json`:

```
{
  "name": "express-app",
  "version": "1.0.0",
  "description": "Simple Express app for Docker",
  "main": "server.js",
  "scripts": {
    "start": "node server.js"
  },
  "dependencies": {
    "express": "^4.18.2"
  }
}
```

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server.js:

```
const express = require('express');
const app = express();
const PORT = 3000;

app.get('/', (req, res) => {
  res.json({ message: 'Hello from Docker Express!' });
});

app.get('/hello/:name', (req, res) => {
  const name = req.params.name;
  res.json({ message: `Hello ${name}!` });
});

app.listen(PORT, '0.0.0.0', () => {
  console.log(`Server running on port ${PORT}`);
});
```

Hints:

Step 1 - Create directory and files:

```
mkdir express-app
cd express-app
# Create package.json and server.js (use the samples above)
```

Step 2 - Write Dockerfile:

```
FROM node:18-alpine
WORKDIR /app
COPY package.json .
RUN npm install
COPY server.js .
EXPOSE 3000
CMD ["node", "server.js"]
```

Step 3 - Build the image:

```
docker build -t express-app:1.0 .
```

Step 4 - Run the container:

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```
docker run -d -p 3000:3000 --name express-container express-app:1.0
```

Step 5 - Test the API:

```
curl http://localhost:3000
```

Output: {"message":"Hello from Docker Express!"}

```
curl http://localhost:3000/hello/Student
```

Output: {"message":"Hello Student!"}